

Chapter 1: Integers

10 worksheets · one for each exercise · Easy, Medium
and Challenging columns

NSW syllabus outcome

Stage 4 Number and algebra: MA4-INT-C-01 compares, orders
and calculates with integers to solve problems.

CHAPTER

1

1.01 Integers and opposites

1.02 The number line

1.03 Comparing and ordering integers

1.04 Adding integers

1.05 Subtracting integers

1.06 Multiplying integers

1.07 Dividing integers

1.08 Order of operations with integers

1.09 Integers on a calculator

1.10 Integer problems

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

WE DO

Top of each column. Your teacher works the examples on the board. **Copy every step** into the space.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

YOU DO

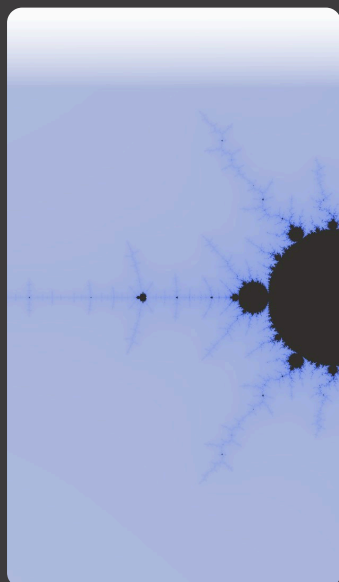
The rest of the column. Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–27

Write the opposite of each integer. It is the same distance from 0, on the other side.

WE DO With your teacher: copy the working.

E1 16

E2 7

YOU DO Your turn.

1 -59 59

2 41

3 9

4 47

5 -25

6 54

7 -14

8 -52

9 -43

10 23

11 27

12 -21

13 -9

14 29

15 -34

16 -2

17 -55

18 57

Write an integer for each situation.

19 63 floors up

20 a fall of 8 points

21 a fall of 48 points

22 50 m below sea level

23 52 minutes late

24 a gain of 4 kg

25 22 m below sea level

26 a loss of \$9

27 48 steps back

Medium

Questions 28–37

Simplify. The opposite of the opposite is the number itself.

WE DO With your teacher: copy the working.

E1 $-(+26)$

E2 $-(-16)$

YOU DO Your turn.

28 $-(+49)$

= the opposite of 49

= -49

29 $-(+57)$

= the opposite of 57

= _____

30 $-(-25)$

= the opposite of _____

= _____

31 $-(-29)$

32 $-(-41)$

33 $-(-50)$

34 $-(-45)$

35 $-(-22)$

36 $-(-32)$

37 $-(-21)$

Challenging

Questions 38–44

Answer each question. Write your reasoning on the lines.

WE DO With your teacher: copy the working.

E1 The opposite of a number is -14. What is the number?

E2 A diver is 7 m below sea level. Write this as an integer, then write its opposite.

YOU DO Your turn.

38 List all the integers whose distance from 0 is less than 3.

from -2 up to 2

_____, _____, _____, _____, _____

39 If $n = -39$, find $-n$ and $-(-n)$.

$-n = -(\text{_____}) = \text{_____}$

$-(-n) = \text{_____}$

40 The opposite of a number is -26. What is the number?

41 List all the integers whose distance from 0 is less than 6.

42 If $n = -3$, find $-n$ and $-(-n)$.

43 The opposite of a number is -6. What is the number?

44 A diver is 3 m below sea level. Write this as an integer, then write its opposite.

Easy

Questions 1–24

Which integer is further to the right on the number line? Write it on the line.



WE DO With your teacher: copy the working.

E1 5 or -4

E2 -2 or -3

YOU DO Your turn.

1 2 or -6

2

2 8 or -10

3 7 or -13

4 1 or -2

5 14 or -4

6 -2 or -11

7 -1 or 11

8 14 or 8

9 -13 or -7

10 3 or -12

11 10 or 1

12 3 or 2

13 15 or -5

14 -11 or 6

15 -11 or 0

16 5 or -15

17 5 or -12

18 -6 or -3

Start at the first number and move along the number line. Where do you land?



19 start at -1, move 7 right

20 start at -5, move 4 left

21 start at 3, move 7 right

22 start at 5, move 2 left

23 start at 5, move 4 left

24 start at 7, move 5 left

Medium

Questions 25–33

How many units apart are the two integers on the number line?



WE DO With your teacher: copy the working.

E1 -7 and 11

E2 -8 and 3

YOU DO Your turn.

25 3 and -1

-1 to 0 is 1, 0 to 3 is 3

1 + 3 = 4

26 11 and -12

-12 to 0 is 12, 0 to 11 is 11

_____ + _____ = _____

27 12 and -13

_____ to _____ is _____, _____ to _____

_____ + _____ = _____

28 5 and -11

29 -9 and 11

30 -14 and 7

Which integer is exactly halfway between?



31 -10 and -6

32 -1 and 7

33 -11 and -3

Challenging

Questions 34–38

Count the integers or find the halfway point. Show your working.



WE DO With your teacher: copy the working.

E1 How many integers are between -8 and 5?

E2 How many integers are between -4 and 23?

YOU DO Your turn.

34 Which integer is halfway between -1 and 25?

distance apart: 26

half: 13

_____ + _____ = _____

35 Which integer is halfway between -13 and 13?

distance apart: _____

half: _____

_____ + _____ = _____

36 How many integers are between -30 and 26?

37 Which integer is halfway between -23 and -1?

38 How many integers are between -27 and 16?

Easy

Questions 1–27

Write $<$ or $>$ in the box. The integer further right is larger.

WE DO With your teacher: copy the working.

E1 $-7 \square -8$

E2 $-5 \square -10$

YOU DO Your turn.

1 $-1 \square -20$

2 $-12 \square 12$

3 $7 \square -11$

4 $2 \square 10$

5 $11 \square 0$

6 $-9 \square -6$

7 $-4 \square -6$

8 $-1 \square -17$

9 $-5 \square -15$

10 $-10 \square -6$

11 $-10 \square 0$

12 $-19 \square 3$

13 $1 \square 11$

14 $5 \square -10$

15 $-4 \square -2$

16 $9 \square -6$

17 $-11 \square 12$

18 $7 \square -19$

Write the smaller integer.

19 -7 or -2

20 2 or -14

21 0 or -14

22 7 or -12

23 -21 or -1

24 6 or -26

25 -24 or -21

26 -14 or 3

27 -30 or -28

Medium

Questions 28–36

Write the integers in ascending order (smallest first).

WE DO With your teacher: copy the working.

E1 $15, -8, -22, -9$

E2 $5, 14, -24, -18$

YOU DO Your turn.

28 $5, -20, 3, 0$

smallest: -20 , largest: 5

$-20, 0, 3, 5$

29 $-24, 5, 15, -6$

smallest: -24 , largest: 15

30 $10, -7, -8, 4$

smallest: _____, largest: _____

31 $-23, -14, 4, 15$

32 $6, -16, 20, -25$

33 $-10, -19, 17, 9$

Write the integers in descending order (largest first).

34 $12, -27, 25, -32$

35 $9, -22, -16, -13$

36 $-39, -23, -31, -2$

Challenging

Questions 37–43

Simplify any $-(-n)$ first, then answer the question.

WE DO With your teacher: copy the working.

E1 Write three integers greater than -14 but less than -9 .

E2 Which is the smallest: $-(-7)$, -7 or $-(-12)$?

YOU DO Your turn.

37 Write in ascending order: $-(-2)$, -9 , -12 , 0

$-(-2) = 2$

38 Write three integers greater than -18 but less than -12 .

integers from _____ to _____

e.g. _____, _____, _____

39 Which is the smallest: $-(-7)$, -8 or $-(-9)$?

40 Write in ascending order: $-(-4)$, -7 , -3 , 0

41 Write three integers greater than -19 but less than -11 .

42 Which is the smallest: $-(-2)$, -6 or $-(-11)$?

43 Write in ascending order: $-(-9)$, -7 , -11 , 0

Easy

Questions 1–27

Evaluate each sum. Start at the first number and move along the number line.



WE DO With your teacher: copy the working.

E1 $-4 + 8$ _____

E2 $-7 + 10$ _____

YOU DO Your turn.

1 $-7 + 1$ -6

2 $-4 + 10$ _____

3 $-4 + 3$ _____

4 $-3 + 1$ _____

5 $-11 + 2$ _____

6 $-2 + 1$ _____

7 $-8 + 10$ _____

8 $-8 + 2$ _____

9 $-10 + 11$ _____

10 $-9 + 5$ _____

11 $-7 + 7$ _____

12 $-11 + 11$ _____

13 $-10 + 7$ _____

14 $-9 + 8$ _____

15 $-5 + 5$ _____

16 $-2 + 8$ _____

17 $-1 + 12$ _____

18 $-8 + 12$ _____

Evaluate. Adding a negative is the same as subtracting.



19 $8 + (-1)$ _____

20 $11 + (-1)$ _____

21 $-6 + (-7)$ _____

22 $10 + (-11)$ _____

23 $-4 + (-12)$ _____

24 $11 + (-4)$ _____

25 $12 + (-2)$ _____

26 $12 + (-3)$ _____

27 $7 + (-1)$ _____

Medium

Questions 28–37



Evaluate. Add from left to right.

WE DO With your teacher: copy the working.

E1 $-12 + (-11) + (-13)$ _____

E2 $-4 + (-3) + (-6)$ _____

YOU DO Your turn.

28 $-4 + 14 + (-5)$

$= 10 + (-5)$

$= 5$

29 $-7 + (-3) + 9$

$= -10 + 9$

$=$ _____

30 $10 + 2 + 5$

$=$ _____ + _____

$=$ _____

31 $-15 + 4 + 1$ _____

32 $3 + (-8) + (-5)$ _____

33 $-13 + 7 + 4$ _____

34 $13 + 9 + 2$ _____

35 $6 + 4 + (-3)$ _____

36 $4 + 6 + (-7)$ _____

37 $-9 + 14 + 1$ _____

Challenging

Questions 38–44

Write a number sentence, then answer the question.



WE DO With your teacher: copy the working.

E1 The temperature is -7°C and rises 8°C . What is the new temperature?

E2 A lift at level -3 goes up 6 floors. Which level is it on?

YOU DO Your turn.

38 Sam has \$35 and spends \$51 on credit. What is the balance?

$35 + (-51)$

$=$ _____

39 Find the missing number: $-12 + \square = 6$

from _____ to _____ is _____ to the right

$\square =$ _____

40 The temperature is -6°C and rises 16°C . What is the new temperature?

41 A lift at level -2 goes up 5 floors. Which level is it on?

42 Sam has \$34 and spends \$62 on credit. What is the balance?

43 Find the missing number: $-5 + \square = 2$

44 The temperature is -12°C and rises 6°C . What is the new temperature?

Easy

Questions 1–27

Evaluate. Subtracting moves you left on the number line.



WE DO With your teacher: copy the working.

E1 $-6 - 4$ _____

E2 $-5 - 4$ _____

YOU DO Your turn.

1 $-8 - 7$ _____ **-15**

2 $-8 - 15$ _____

3 $2 - 2$ _____

4 $5 - 15$ _____

5 $5 - 13$ _____

6 $9 - 12$ _____

7 $6 - 10$ _____

8 $1 - 3$ _____

9 $-3 - 3$ _____

10 $0 - 8$ _____

11 $0 - 10$ _____

12 $8 - 2$ _____

13 $-8 - 10$ _____

14 $-6 - 3$ _____

15 $-1 - 9$ _____

16 $6 - 13$ _____

17 $4 - 3$ _____

18 $-5 - 10$ _____

Evaluate. Subtracting a negative is the same as adding.



19 $12 - (-9)$ _____

20 $-2 - (-11)$ _____

21 $10 - (-1)$ _____

22 $8 - (-8)$ _____

23 $2 - (-9)$ _____

24 $-12 - (-5)$ _____

25 $3 - (-3)$ _____

26 $11 - (-10)$ _____

27 $-4 - (-8)$ _____

Medium

Questions 28–34

Evaluate. Change each $- (-)$ to $+$, then add from left to right.

WE DO With your teacher: copy the working.

E1 $1 - (-15) + (-15)$ _____

E2 $7 - (-13) + (-6)$ _____

YOU DO Your turn.

28 $-13 - (-7) + (-7)$ _____

$= -13 + 7 + (-7)$

$= -6 + (-7)$

$= -13$

29 $-5 - (-1) + 15$ _____

$= -5 + 1 + 15$

$= -4 + 15$

$= 11$

30 $14 - (-10) + 11$ _____

$= 14 + 10 + 11$

$= 35$

$= 35$

31 $13 - (-7) + (-5)$ _____

32 $6 - (-9) + (-5)$ _____

33 $9 - (-7) + (-7)$ _____

34 $6 - (-6) + (-2)$ _____

Challenging

Questions 35–40

Write a number sentence, then answer the question.



WE DO With your teacher: copy the working.

E1 The minimum temperature was -5°C and the maximum was 19°C . By how much did it rise?

E2 A submarine at -62 m rises 22 m . Where is it now?

YOU DO Your turn.

35 What is 2 less than -9 ?

$-9 - 2$

$= -11$

36 Find the difference between -6 m and 24 m .

$24 - (-6)$

$= 24 + 6$

$= 30\text{ m}$

37 The minimum temperature was -10°C and the maximum was 11°C . By how much did it rise?

38 A submarine at -105 m rises 22 m . Where is it now?

39 What is 9 less than -4 ?

40 Find the difference between -15 m and 18 m .

Easy

Questions 1–27

Evaluate each product.
Same signs: positive.
Different signs: negative.

WE DO With your teacher: copy the working.

E1 -5×3

E2 $6 \times (-2)$

YOU DO Your turn.

1 $-7 \times (-9)$

63

2 -7×7

3 $-3 \times (-1)$

4 $1 \times (-3)$

5 $-1 \times (-7)$

6 $8 \times (-9)$

7 $4 \times (-9)$

8 -8×8

9 -10×8

10 $5 \times (-1)$

11 $-7 \times (-8)$

12 -4×9

13 $-8 \times (-5)$

14 -8×7

15 $-3 \times (-10)$

16 $8 \times (-3)$

17 $6 \times (-5)$

18 $-2 \times (-2)$

Write the sign of the answer only: + or -.

19 $2 \times (-7)$

20 $(-1) \times 9$

21 3×1

22 2×5

23 $(-2) \times 1$

24 $(-3) \times 3 \times 4$

25 7×7

26 $4 \times (-3)$

27 $1 \times (-1) \times 3$

Medium

Questions 28–47

Evaluate. Multiply two numbers at a time.

WE DO With your teacher: copy the working.

E1 $4 \times (-5) \times (-3)$

E2 $1 \times (-1) \times 4$

YOU DO Your turn.

28 $-6 \times (-2) \times (-3)$

$= 12 \times (-3)$

$= -36$

30 $-6 \times 5 \times (-5)$

$= ____ \times (____)$

$= ____$

32 $-3 \times 4 \times 5$

34 $-3 \times 3 \times 2$

36 $5 \times (-5) \times 4$

38 $5 \times 4 \times (-5)$

40 $-4 \times (-1) \times 4$

42 $-4 \times 6 \times (-1)$

44 $2 \times (-4) \times 1$

46 $-4 \times 1 \times 4$

29 $5 \times (-1) \times (-2)$

$= -5 \times (-2)$

$= ____$

31 $-1 \times (-1) \times (-1)$

33 $3 \times (-4) \times 1$

35 $-1 \times 1 \times 2$

37 $-3 \times 6 \times (-1)$

39 $5 \times (-4) \times 1$

41 $-5 \times 1 \times (-1)$

43 $-1 \times (-5) \times (-3)$

45 $-6 \times 5 \times (-4)$

47 $-6 \times (-2) \times 2$

Challenging

Questions 48–53

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A diver goes down 2 m every minute. Where is she after 3 minutes?

E2 The temperature drops 4°C each hour for 3 hours. What is the total change?

YOU DO Your turn.

48 Find the missing number: $-4 \times \square = -12$

$\square = -12 \div (-4)$

$\square = ____$

49 A diver goes down 2 m every minute. Where is she after 6 minutes?

$____ \times ____$

$= ____ \text{ m}$

50 The temperature drops 6°C each hour for 6 hours. What is the total change?

51 Find the missing number: $-4 \times \square = 32$

52 A diver goes down 4 m every minute. Where is she after 7 minutes?

53 The temperature drops 5°C each hour for 9 hours. What is the total change?

Easy

Questions 1–27

Evaluate each quotient.
Same signs: positive.
Different signs: negative.



WE DO With your teacher: copy the working.

E1 $36 \div (-4)$ _____

E2 $-28 \div (-7)$ _____

YOU DO Your turn.

1 $8 \div (-8)$ _____

-1

2 $90 \div (-9)$ _____

3 $-4 \div (-1)$ _____

4 $40 \div (-8)$ _____

5 $4 \div (-1)$ _____

6 $-40 \div (-5)$ _____

7 $-56 \div 7$ _____

8 $40 \div (-5)$ _____

9 $-30 \div (-10)$ _____

10 $-12 \div 3$ _____

11 $-4 \div (-4)$ _____

12 $-10 \div 1$ _____

13 $-24 \div 6$ _____

14 $15 \div (-5)$ _____

15 $15 \div (-3)$ _____

16 $-4 \div 4$ _____

17 $-90 \div 9$ _____

18 $30 \div (-3)$ _____

Find the missing number.



19 $30 \div \square = -5$ _____

20 $36 \div \square = 4$ _____

21 $8 \div \square = 8$ _____

22 $25 \div \square = 5$ _____

23 $-14 \div \square = 7$ _____

24 $-5 \div \square = 5$ _____

25 $-5 \div \square = -5$ _____

26 $-9 \div \square = 9$ _____

27 $-20 \div \square = 4$ _____

Medium

Questions 28–47

Evaluate. Divide from left to right.

WE DO With your teacher: copy the working.

E1 $8 \div 4 \div 2$ _____

E2 $-40 \div (-5) \div (-2)$ _____

YOU DO Your turn.

28 $-10 \div 1 \div 2$ _____

$= -10 \div 2$ _____

$= -5$ _____

30 $8 \div (-1) \div (-4)$ _____

$= ______ \div (______)$ _____

$= ______$ _____

32 $-8 \div 2 \div (-2)$ _____

29 $-9 \div 1 \div 3$ _____

$= -9 \div 3$ _____

$= ______$ _____

31 $-8 \div 2 \div 4$ _____

33 $-15 \div (-5) \div 3$ _____

34 $-50 \div (-5) \div (-2)$ _____

35 $-32 \div (-4) \div (-4)$ _____

36 $45 \div 5 \div 3$ _____

37 $10 \div 1 \div (-2)$ _____

38 $4 \div 4 \div 1$ _____

39 $32 \div (-4) \div 4$ _____

40 $16 \div 4 \div 2$ _____

41 $-60 \div 3 \div (-4)$ _____

42 $36 \div 3 \div 4$ _____

43 $6 \div (-2) \div 1$ _____

44 $24 \div (-3) \div (-4)$ _____

45 $-6 \div (-1) \div 3$ _____

46 $-9 \div 3 \div 1$ _____

47 $50 \div 5 \div (-2)$ _____

Challenging

Questions 48–53

Write a number sentence, then answer the question.



WE DO With your teacher: copy the working.

E1 A debt of \$160 is shared equally by 8 people. Write each share as an integer.

E2 The temperature fell 15°C over 3 hours. What was the change each hour?

YOU DO Your turn.

48 Find the missing number: $\square \div 8 = 4$

$\square = 4 \times 8$ _____

$\square = ______$ _____

49 A debt of \$114 is shared equally by 6 people. Write each share as an integer.

$______ \div ______$ _____

$= ______ \text{ each}$ _____

50 The temperature fell 12°C over 6 hours. What was the change each hour?

51 Find the missing number: $\square \div (-4) = -5$

52 A debt of \$126 is shared equally by 7 people. Write each share as an integer.

53 The temperature fell 24°C over 4 hours. What was the change each hour?

Easy

Questions 1–27

Evaluate. $\times$ and $\div$ before $+$ and $-$.



WE DO With your teacher: copy the working.

E1 $3 + (-6) \times (-6)$ _____

E2 $-9 - 5 \times (-6)$ _____

YOU DO Your turn.

1 $-5 - (-6) \times (-4)$ _____

-29

2 $-12 - (-2) \times (-2)$ _____

3 $-6 - (-5) \times (-6)$ _____

4 $10 + 6 \times 4$ _____

5 $2 + 4 \times 5$ _____

6 $-3 - (-4) \times 4$ _____

7 $7 + (-6) \times (-2)$ _____

8 $9 - 2 \times (-2)$ _____

9 $3 - (-4) \times 4$ _____

10 $9 + 4 \times 2$ _____

11 $8 + 5 \times (-6)$ _____

12 $-10 + 2 \times (-6)$ _____

13 $-4 - 2 \times (-6)$ _____

14 $8 + (-1) \times (-3)$ _____

15 $1 + 3 \times (-1)$ _____

16 $-2 - 5 \times 5$ _____

17 $-6 - 4 \times 2$ _____

18 $5 - (-1) \times 5$ _____

Evaluate. Brackets first.



19 $(-6 - (-1)) \times 2$ _____

20 $(2 + (-9)) \times 3$ _____

21 $(-5 - 7) \times 4$ _____

22 $(6 - 3) \times 3$ _____

23 $(2 + (-4)) \times 1$ _____

24 $(4 - (-9)) \times 1$ _____

25 $(7 - (-1)) \times (-4)$ _____

26 $(-5 - (-6)) \times 5$ _____

27 $(-6 + 4) \times (-2)$ _____

Medium

Questions 28–37

Evaluate. Do $\times$ and $\div$ first (left to right), then $+$ and $-$.

WE DO With your teacher: copy the working.

E1 $6 \times 1 + 2 \div 1$ _____

E2 $3 \times (-6) + (-16) \div 4$ _____

YOU DO Your turn.

28 $-5 \times (-6) - (-5) \div (-5)$ _____

= 30 - 1

= 29

29 $-1 \times (-2) + (-10) \div 2$ _____

= 2 + (-5)

= -3

30 $1 \times 6 + (-10) \div 2$ _____

= 6 + (-5)

= 1

31 $1 \times 3 + 6 \div (-3)$ _____

32 $4 \times 1 + 15 \div 5$ _____

33 $6 \times (-4) - 8 \div 4$ _____

34 $6 \times (-2) - 20 \div 4$ _____

35 $-5 \times (-3) + 5 \div (-5)$ _____

36 $-6 \times 6 - 6 \div (-3)$ _____

37 $3 \times (-4) - 16 \div (-4)$ _____

Challenging

Questions 38–44

Evaluate. Brackets first, then $\times$ and $\div$, then $+$ and $-$. One step on each line.



WE DO With your teacher: copy the working.

E1 $7 - (-4 + 6) \times (-3)$ _____

E2 $6 - (-1 + (-2)) \times (-1)$ _____

YOU DO Your turn.

38 $-5 - (7 + 6) \times (-3)$ _____

= -5 - 13 \times (-3)

= -5 - (-39)

= 34

39 $5 - (2 + 8) \times (-1)$ _____

= 5 - 10 \times (-1)

= 5 - (-10)

= 15

40 $-5 - (5 + 8) \times (-3)$ _____

41 $2 - (7 + (-4)) \times (-4)$ _____

42 $7 - (8 + (-1)) \times 3$ _____

43 $-7 - (-4 + 5) \times (-3)$ _____

44 $4 - (7 + 7) \times (-4)$ _____

Easy

Questions 1–27

Use your calculator. Use the (–) key for negative numbers.



WE DO With your teacher: copy the working.

E1 $252 + (-391)$ _____

E2 $-369 + 350$ _____

YOU DO Your turn.

1 $27 + 17$ _____ **44**

2 $-160 + (-124)$ _____

3 $24 + 106$ _____

4 $-198 + 77$ _____

5 $-269 + 255$ _____

6 $269 + (-274)$ _____

7 $267 + 146$ _____

8 $-158 + 224$ _____

9 $-396 + 241$ _____

10 $197 + (-106)$ _____

11 $59 + (-266)$ _____

12 $-303 + (-27)$ _____

13 $296 + 12$ _____

14 $224 + 100$ _____

15 $127 + 343$ _____

16 $-299 + 318$ _____

17 $254 + 3$ _____

18 $-72 + 281$ _____

Use your calculator.



19 $43 \times (-4)$ _____

20 22×3 _____

21 $45 \times (-44)$ _____

22 $3 \times (-11)$ _____

23 $-45 \times (-20)$ _____

24 -22×12 _____

25 $17 \times (-33)$ _____

26 $31 \times (-32)$ _____

27 $13 \times (-33)$ _____

Medium

Questions 28–37

Use your calculator. Enter the brackets, then check the answer makes sense.



WE DO With your teacher: copy the working.

E1 $(-86 - (-81)) \times (-16)$ _____

E2 $(-3 - 67) \times (-11)$ _____

YOU DO Your turn.

28 $(-86 - (-8)) \times (-22)$ _____

bracket: -78

$-78 \times (-22) = 1\,716$

29 $(12 - 22) \times 19$ _____

bracket: -10

_____ $\times$ _____ = _____

30 $(12 - (-59)) \times (-18)$ _____

bracket: _____

_____ $\times$ (_____) = _____

31 $(-44 - 62) \times (-21)$ _____

32 $(45 - 80) \times (-4)$ _____

33 $(-90 - 59) \times (-19)$ _____

34 $(-31 - 52) \times (-15)$ _____

35 $(57 - 83) \times 15$ _____

36 $(-74 - (-90)) \times (-3)$ _____

37 $(-62 - 84) \times (-15)$ _____

Challenging

Questions 38–44

Estimate first, then use your calculator. Is your answer close to the estimate?



WE DO With your teacher: copy the working.

E1 Estimate -60×44 , then calculate.

E2 Estimate -32×48 , then calculate.

YOU DO Your turn.

38 Estimate -34×17 , then calculate.

estimate: $-30 \times 20 \approx -600$

calculator: _____

39 Estimate -58×36 , then calculate.

estimate: _____ $\times$ _____ $\approx$ _____

calculator: _____

40 Estimate -42×35 , then calculate.

41 Estimate -46×28 , then calculate.

42 Estimate -38×30 , then calculate.

43 Estimate -57×13 , then calculate.

44 Estimate -48×26 , then calculate.

Easy

Questions 1–27

Find the new temperature.

WE DO With your teacher: copy the working.

E1 5°C, rises 14°C

E2 14°C, falls 4°C

YOU DO Your turn.

1 -4°C, falls 17°C

-21°C

2 -7°C, falls 16°C

3 -9°C, rises 15°C

4 11°C, falls 7°C

5 7°C, falls 5°C

6 -10°C, rises 10°C

7 5°C, falls 11°C

8 -2°C, falls 15°C

9 11°C, rises 10°C

10 12°C, falls 13°C

11 -2°C, falls 12°C

12 13°C, falls 17°C

13 -10°C, falls 14°C

14 13°C, rises 13°C

15 8°C, rises 4°C

16 -6°C, falls 3°C

17 14°C, rises 15°C

18 -9°C, rises 12°C

Find the new bank balance.

19 -\$72, deposit \$92

20 -\$9, withdraw \$103

21 -\$23, deposit \$24

22 \$133, deposit \$70

23 \$118, deposit \$53

24 -\$73, withdraw \$18

25 \$103, withdraw \$78

26 -\$9, deposit \$86

27 \$13, deposit \$42

Medium

Questions 28–34

Find the difference between the two heights. Write a number sentence.

WE DO With your teacher: copy the working.

E1 a hill 267 m above sea level and a reef 57 m below

E2 a hill 280 m above sea level and a reef 74 m below

YOU DO Your turn.

28 a hill 106 m above sea level and a reef 10 m below

$$106 - (-10)$$

$$= 116 \text{ m}$$

29 a hill 66 m above sea level and a reef 19 m below

$$66 - (-19)$$

$$= \text{ } \text{ m}$$

30 a hill 52 m above sea level and a reef 32 m below

$$\text{ } - (\text{ })$$

$$= \text{ } \text{ m}$$

31 a hill 174 m above sea level and a reef 48 m below

32 a hill 388 m above sea level and a reef 57 m below

33 a hill 129 m above sea level and a reef 45 m below

34 a hill 28 m above sea level and a reef 88 m below

Challenging

Questions 35–38

Write a number sentence for each step, then answer the question.

WE DO With your teacher: copy the working.

E1 A diver is at -45 m. She rises 4 m a minute for 4 minutes. Where is she now?

E2 A game scores +4 for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score?

YOU DO Your turn.

35 It is -2°C at 6 a.m. It warms 3°C an hour for 3 hours. What is the temperature?

$$3 \times 3 = 9$$

$$-2 + 9$$

$$= \text{ } ^\circ\text{C}$$

36 A diver is at -30 m. She rises 5 m a minute for 5 minutes. Where is she now?

$$\text{ } \times \text{ } = \text{ }$$

$$\text{ } + \text{ }$$

$$= \text{ } \text{ m}$$

37 A game scores +4 for a win and -1 for a loss. Mia wins 4 and loses 4. What is her score?

38 It is -3°C at 6 a.m. It warms 5°C an hour for 6 hours. What is the temperature?